Honest Evaluation of Underwater Acoustic Target Recognition: Decomposing, Predicting, and Reducing Benchmark Variance

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Abstract

Identity-disjoint evaluation is now the recommended convention in underwater acoustic target recognition (UATR), but its statistical cost has never been quantified. On OceanShip, the vessel-disjoint accuracy of an identical model varies by ± 5.0 pp across 25 independent draws—twelve times the clip-level noise—while the clip-vs-honest gap is 5.1 pp on OceanShip but 30.4 pp on DeepShip. This draw-to-draw variance is dominated (89% in-sample; 59% out-of-sample under leave-one-draw-out) by the

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All frozen splits, allocation code, per-clip predictions, and the required-N calculator are released for replication.

class composition of the test set, computable from metadata alone; a first-order predictor calibrated on it correctly estimated the observed spread of five allocation schemes across three model families; and a class-wise clip-mass stratified allocation collapses the variance to the lowest level observed across the tested allocations (4.69 to 1.24 pp; variance ratio 14.4, exact permutation $p = 0.004$), reducing the draws needed to resolve a 2 pp improvement from about 100 to 8 per system (10–13 to 3–7 when comparisons are fully paired). Across models, the remedy shows no benefit on DeepShip, where composition-induced variability is small relative to the observed draw variability. In the tested settings and diagnostics, no pipeline defect emerged behind the low absolute accuracy of honest evaluation: five tested interventions—capacity growth, post-hoc prior adjustment, selective prediction, frozen pretraining, and full fine-tuning—fail to decode the nine-class tail, and a training-mass diagnostic shows recall tracks clip identity mass; the tail's three-tier structure is itself predicted from metadata. Frozen splits, allocation code, and a required-N calculator are released.

Index Terms

Underwater acoustic target recognition, vessel-disjoint evaluation, benchmark variance, group-stratified allocation, data leakage.

I. Introduction

Recent guidance in underwater acoustic target recognition (UATR) has converged on a simple methodological prescription: do not evaluate with clip-level random splits. Adjacent clips of one recording share background noise and propagation conditions [1], so a random clip split places near-duplicates on both sides of the train/test boundary, and the resulting accuracy measures retrieval of a known recording as much as recognition of a ship type. The 2026 survey of the field lists the accuracy gaps between random, temporal, and recording-level splits explicitly [1]; UniqueShip [2] re-quantifies the inflation at 10–48 pp across corpora and proposes identity-grouped re-splits; and recent systems work adopts recording-level splits with significance testing [3]. Within the general machine-learning literature, leakage taxonomies [4] and variance accounting [5] provide the conceptual vocabulary.

This paper starts from the observation that the guidance leaves out: *honest evaluation has a noise floor, and, to our knowledge, no one has measured it.* When the unit of allocation is the vessel or the recording, each random draw changes which ships—and therefore which classes—land in the test set. On OceanShip, a public benchmark with vessel identities, the test accuracy of an identical model under identical training varies by ± 5.0 pp from one vessel-disjoint draw

to the next (25 independent draws), while the clip-random baseline varies by ± 0.4 pp. The honest protocol is therefore simultaneously more meaningful and twelve times noisier; a paper reporting a single honest-protocol number is reporting an estimate whose uncertainty exceeds most published method improvements. And because the noise is driven by *which classes* the draw samples, it is not statistical background: it is a property the benchmark's own metadata determines and exposes.

We make four contributions.

1) **A variance decomposition of honest evaluation** (Sec. V). We split the draw-to-draw variance into a *composition* component (which classes the draw places in the test set; pure metadata, computable before training) and a residual component (initialization, optimization, and any remaining draw-level effects). On OceanShip the composition component explains 89% of the vessel-disjoint variance over 25 independent draws (permutation $p = 0.0006$), 54% at the intermediate recording level, and 0% at the clip level: the more honest the protocol, the more the score is determined by metadata ($0\% \to 54\% \to 89\%$).
2) **A validated, calibration-based predictor** (Sec. V-C). Fed with one pooled per-class recall table (measured once from trained models—scoring any candidate allocation requires no retraining) and the splits' class composition, the model predicted the observed draw spread of five independent allocation schemes across three model families, and correctly predicted the *absence* of fixable variance on DeepShip (Sec. VI-D).
3) **A remedy with measured variance reduction** (Sec. VI). A class-wise clip-mass stratified allocation—assigning whole vessels until each class has contributed its quota of *clips*—collapses the honest-protocol variance to the lowest level observed across the tested allocations (4.69 pp $\to 1.24$ pp; variance ratio 14.4, exact permutation $p = 0.004$), generalizes across a linear probe and a $4\times$ wider CNN, and reduces the draws needed to resolve a 2 pp improvement from about 100 independent draws to 8—and paired two-model comparisons need 3–7 draws rather than 10–13. A naive stratification balancing *vessel counts* instead of clip masses fails exactly as the decomposition predicts.
4) **The limits of tested remedies** (Sec. VII). In the tested settings and diagnostics, no pipeline defect emerged behind the low absolute accuracy of honest evaluation: five tested interventions—capacity growth, post-hoc prior adjustment, selective prediction, frozen pretrained encoders, and full fine-tuning—fail to decode the nine-class tail, a training-mass diagnostic explains part of the pattern, and the three-tier structure (learnable, partially

rescuable, intervention-resistant) is predictable from metadata.

Throughout, we separate two quantities that the leakage literature tends to conflate: the *existence* of leaky information (provable from metadata: on OceanShip, predicting a test clip's label from the majority label of its training vessel reaches 99.4%) and its *acoustic exploitability* (a nearest-neighbour probe on the features reaches only 25.5%, with same-recording neighbours at 5.7% versus a 0.03% chance rate). On DeepShip, where recordings are tiled into contiguous 5 s windows, a single scalar—the source-file duration, one value per recording—retrieves 78.2% of clip-random accuracy without listening to the audio. Existence without exploitability, and exploitability without existence, are different failure modes with different remedies; we measure both (Sec. IV).

II. Related Work

A. Leakage in ML-Based Science

Kapoor and Narayanan [4] enumerate leakage types across disciplines; duplicate-adjacent and non-independent train/test samples (their L1.4/L3.2) are the UATR case. Recent work formalizes the distinction we exploit: metadata predictability of an output (MPDS) is logically independent of the output's dependence on the underlying evidence [6]. In audio specifically, mel-similarity audits have surfaced tens of millions of overlapping pairs across popular datasets, and UniqueShip [2] re-quantifies leakage for acoustic ship classification, reporting 10–48 pp inflation under random splits and proposing identity-grouped re-splits. Our work is complementary: we take identity grouping as given and ask what it does to the *reliability* of the resulting evaluation.

B. Evaluation Noise and Variance

Bouthillier et al. [5] showed that data-sampling variance dominates initialization variance in general ML benchmarks; Gillick and Cox [7] established matched-pairs significance testing for speech evaluation—the same paired-across-draws design we use throughout; recent UATR guidance discusses grouped evaluation at the recording level [1]. None of these quantify *which* component of group-disjoint variance is metadata-predictable, nor the number of draws required for a given effect size—the two numbers a practitioner actually needs.

C. Group-Stratified Splitting

Stratified group k-fold [8] balances class margins across folds while keeping groups intact, and count- or margin-balanced group variants are standard tooling. We do not claim a new algorithm:

our clip-mass allocator is a quota variant of the same idea, distinguished by (i) balancing the per-class *clip mass* of vessels rather than their counts—a distinction the data proves matters (Sec. VI)—and (ii) being validated by a pre-specified, decomposable variance model rather than offered as a heuristic; none of the standard variants ships with a variance model that says when balancing matters at all.

D. UATR Benchmarks and Systems

DeepShip [9] and ShipsEar [10] tile or excerpt recordings into labeled clips; OceanShip [11] adds vessel identities (MMSI) and AIS metadata. Systems work adopts recording-level splits with statistical checks [3]; transfer-learning studies report that embedding spaces are dominated by recording-specific factors [12]; open-set recognition, contrastive generalization [13], adversarial individual-vessel generalization [14], and physics-informed architectures [15] target the accuracy race under various protocols. Our audit explains a cross-cutting observation in this literature: under honest evaluation, 12-class OceanShip collapses to three effective classes while 4-class DeepShip does not (Sec. VII)—a consequence of class granularity that no accuracy number reveals by itself.

III. Data, Protocols, and Models

A. Corpora

OceanShip [11] is the primary benchmark: 53,770 clips from 10,168 hydrophone recordings of 892 vessels (MMSI), labeled by AIS ship type, 5 s clips at 32 kHz. We evaluate the 12 classes with at least 3 vessels; the 3 rare classes (Diving ship, Military ship, Spare; 139 clips) are excluded from all splits, matching how the frozen baseline splits define the task. Recordings are arbitrary-length excerpts of a hydrophone stream: the median recording contains 3 clips (maximum 210), and the trailing index of a clip id is *not* a stream position—two clips of one recording are arbitrary moments, not adjacent windows. **DeepShip** [9] (47 h, 609 recordings, 4 classes) is the structural opposite: recordings are tiled by back-to-back 5 s windows (median 51 clips per recording; clips $\times\ 5\,\mathrm{s}/\text{duration} \approx 0.99$; the clip index equals stream position for 609/609 recordings). **ShipsEar** [10] (2,223 clips, 11 classes) is used only for leakage statistics. Table I contrasts the corpora; this slicing policy, as we show in Sec. IV, is the hidden variable that makes "random split" incomparable across papers.

TABLE I

CORPORA AND SLICING POLICY

	OceanShip	DeepShip
recordings	10,168	609
clips	53,770	33,770
classes (evaluated)	12 (of 15)	4
clips / recording (median)	3	51
tiling ratio (clips$\times$5 s / duration)	n/a	0.994
clip index = stream position	no	609/609
test clips with adjacent training clip	n/a (index is not stream position)	89.6%
acoustic 1-NN nearest training clip from same recording	5.7%	66.8%
clip $-$ honest gap	$+5.1$ pp	$+30.4$ pp

B. Split Protocols

For OceanShip we use three allocation units: **clip random** (clips assigned independently; the leaky convention), **recording random** (recordings intact; ships still cross sets), and **vessel disjoint** (no MMSI on both sides; the honest protocol). For DeepShip the analogues are clip random, recording random, and temporal. All splits are frozen files generated by a deterministic SHA-256-seeded procedure (independent of Python's salted `hash`), verified for zero group intersection at write time and re-verified by the training runner at read time; split draws and network initialization use separate seeds. Protocol names are never trusted: the runner rejects a split file that does not contain the requested (protocol, seed) pair, and a stalled schedule aborts the run. Fig. 1 shows what each policy lets cross the boundary.

C. Models and Training Recipe

The reference model is `cnn_small` (136,692 parameters) trained on log-mel spectrograms (128 mels $\times$ 313 frames) for 24 epochs with AdamW (lr 3×10^{-3}, 3 warm-up epochs, cosine decay, batch 64). The recipe was fixed by a wiring check (the model memorizes 512 clips to 99.2% in 200 epochs; a ResNet-18 memorizes the full training set to 0.9999) and never changed across any table in this paper; every run logs the final learning rate, and a guard aborts any run whose schedule never stepped—two silent failures we encountered and eliminated early. Model families for the remedy experiments: `logreg` (multinomial logistic regression on the flattened

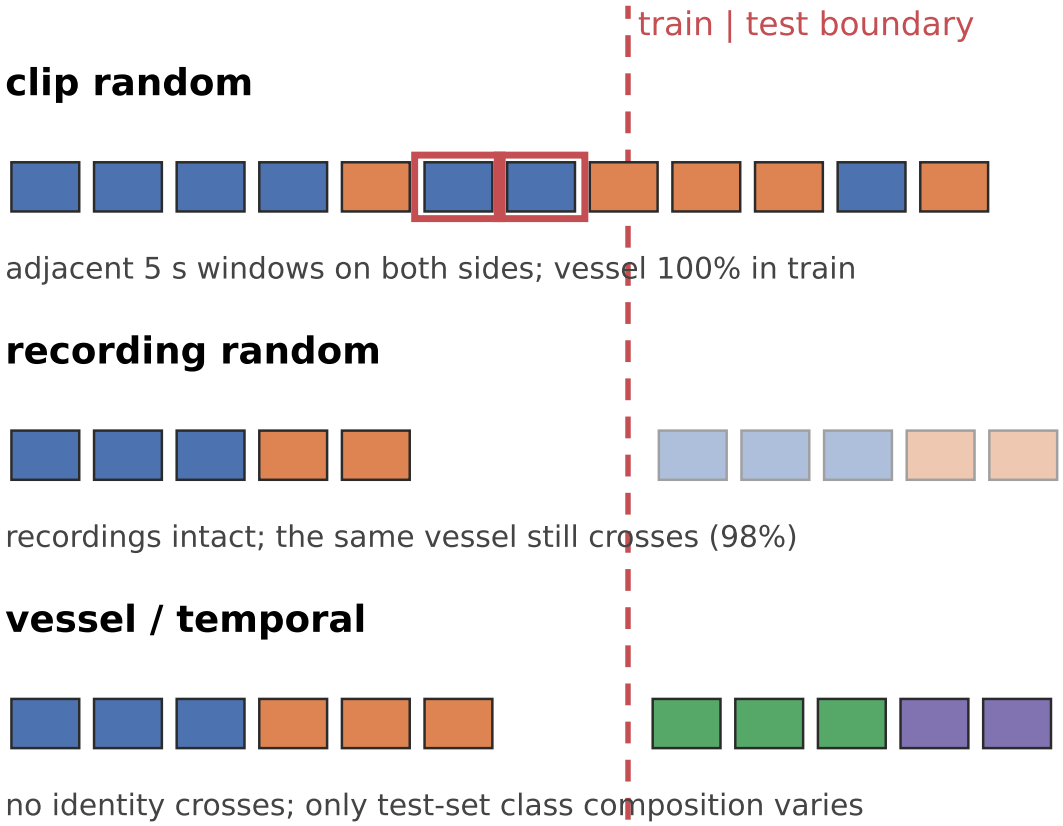


Fig. 1. What crosses the split boundary under each slicing policy (one row of squares = the clips of one or two vessels). Clip random puts adjacent 5 s windows of the *same audio event* on both sides and the vessel's identity in the training set; recording random keeps recordings intact but still lets 98% of vessels cross; vessel/temporal isolation crosses nothing—and thereby turns test-set class composition into a random variable, the subject of Sec. V.

spectrogram; 480,780 parameters) and `cnn_wide` (542,940 parameters). Each run stores per-clip predictions with recording/vessel identity, enabling recording-level bootstrap and per-class decomposition. Table II lists the full recipe; it is intentionally modest, because every claim in this paper is about the evaluation protocol, not about pushing accuracy—and the interventions in Sec. VII bound how much model choice could change our conclusions.

D. Pre-Registration Protocol

Every comparison in this paper was registered before its runs: the verdict thresholds (e.g., the remedy succeeds iff the SD ratio is ≤ 0.6; the fine-tuning door opens iff tail top-2 ≥ 0.30), the control predictions, and the identity-floor prediction for the < 5-vessel classes were

TABLE II

TRAINING RECIPE (IDENTICAL FOR EVERY RUN IN TABLES II–VI)

input	log-mel 128×313, per-clip standardized
optimizer	AdamW, batch 64, weight decay 10^{-4}
schedule	cosine, 3 warm-up epochs, peak lr 3×10^{-3}
epochs	24 (200 for the 512-clip wiring check)
guards	per-epoch lr log; abort if schedule never steps
model selection	final epoch (val used only for fine-tuned AST)
logging	per-clip predictions with recording/vessel id
seeds	split draw and initialization decoupled

written down before the corresponding GPU jobs. Where a prediction failed—our first remedy candidate, weighting by vessel *counts*, was rejected by a zero-cost dose–response check before any training—we report it (Sec. VI). We consider this discipline part of the contribution: the audit's claims are exactly as strong as their pre-registration.

IV. LEAKAGE: EXISTENCE VERSUS EXPLOITABILITY

A. Metadata Ceilings

The metadata ceiling—predicting each test clip's label by the majority training label of its vessel, using no audio—is 99.44% on OceanShip clip-random (99.5% coverage) and 98.16% at the recording level; on DeepShip clip-random the recording ceiling is 99.82%. These numbers quantify *existence*: what the allocation hands over. Labels themselves are clean—no OceanShip recording is multi-labeled, and only 5 of 892 vessels are cross-class (1.1% of clips)—so identity lookup is a legitimate oracle, not label noise.

B. Acoustic Exploitability

Existence does not imply exploitability. On 16×16 pooled log-mel descriptors, the 1-NN probe finds a same-recording nearest training clip for only 5.7% of OceanShip test clips—$180\times$ above the 0.03% chance rate, so recording identity *is* in the features—but weak and diffuse; 1-NN label accuracy is 25.5% against a 22.1% constant baseline, and a ridge probe on the 256-D descriptor reaches 27.4%. On DeepShip the picture inverts: 1-NN label accuracy is 83.2% (27.2% constant), same-recording neighbours at 66.8% ($234\times$ chance), and 36.8% of them are the immediately adjacent 5 s window.

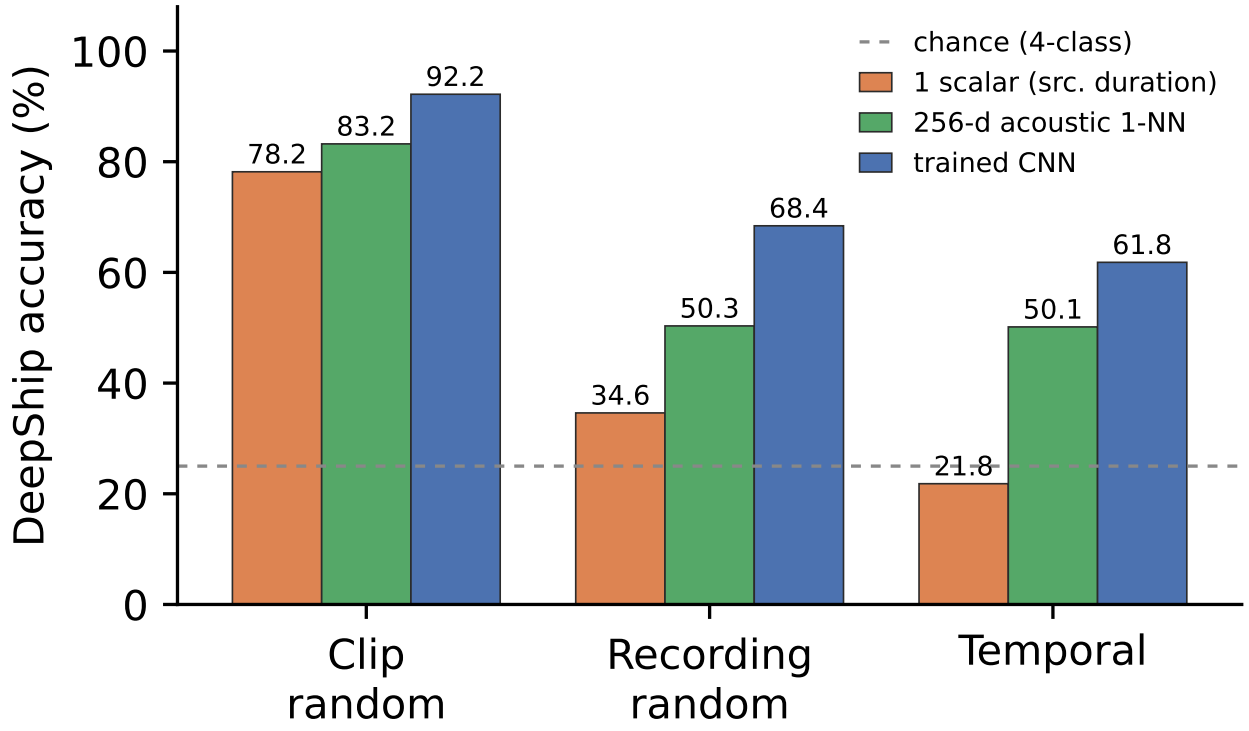


Fig. 2. The one-scalar probe on DeepShip (seed 0): accuracy of a single metadata scalar (source-file duration 1-NN), a 256-d acoustic descriptor 1-NN, and the trained CNN under the three slicing policies. The scalar's 78.2% under clip random buys *identity*, not ship type—it collapses to near chance once recordings are isolated, while the acoustic probes lose far less. Data from the released recording-structure audit.

C. One Scalar Is Enough—Where Recordings Are Tiled

DeepShip's `src_dur_s` column is the *source-file duration*, one value shared by every 5 s window cut from that file. It is therefore a recording fingerprint. A nearest-neighbour classifier on this *single scalar*—no audio, no training—reaches 78.2% (five draws: 78.2–80.0%) on clip random, compared with 83.2% for a 256-D acoustic descriptor and 92.2% for the trained model. The ordering is the finding. Under recording-level splits the scalar collapses to 34.6% (recording random) and 21.8% (temporal), at or below the constant baseline: what the scalar buys is *identity*, not ship type. On OceanShip the same scalar reaches only 9.3%, because clip duration is not a recording fingerprint there. Table III summarizes and Fig. 2 shows the ladder: under clip random, a metadata scalar with no audio nearly matches an acoustic descriptor and a trained CNN; under honest splits every identity-linked advantage collapses toward chance, exposing what the leaky number was made of.

TABLE III

LEAKAGE: EXISTENCE VS. EXPLOITABILITY (SEED-0 DRAWS, NO AUDIO FOR CEILINGS/SCALAR)

	DeepShip	OceanShip
metadata ceiling (vessel), clip random	97.5%	99.4%
metadata ceiling (recording), clip random	99.8%	90.8%
1-NN label accuracy	83.2%	25.5%
(constant baseline)	(27.2%)	(22.1%)
1-NN shares the recording	66.8%	5.7%
(chance rate)	(0.29%)	(0.03%)
adjacent-window neighbours	36.8%	n/a[†]
one-scalar (source duration) 1-NN	78.2%	9.3%
trained model, clip random	92.2%	36.7%

[†]OceanShip carries no stream position; the absence itself is a slicing-policy fact.

The official OceanShip split quantifies the stakes. In the benchmark authors' own configuration, 91.8% of vessels and 19.8% of recordings appear on both sides of the split; our clip-random reconstruction reaches 99.6% and 85.0% of test recordings respectively (the 85.0/90.8/91.1% figures quoted in this paper use different units—by recording vs. by clip—and must be stated when quoted). Against these baselines, the vessel-disjoint protocol is not a refinement but a category change; everything in Secs. V–VI-D is about making that category change usable.

V. THE ANATOMY OF HONEST-EVALUATION VARIANCE

A. The Same Protocol Name Is Not a Specification

Table IV and Fig. 3 give the ladder. On OceanShip, the same model and recipe score 0.367 under clip random and 0.316 under vessel disjoint: the paired leak is $+5.1$ pp (draw SD of the paired difference 4.5 pp, paired t $p = 0.064$)—at five draws the leak itself sits at the edge of significance, which is the noise-floor point of this section, not a weakness of the test. On DeepShip the same nominal comparison spans $0.922 \rightarrow 0.618$: 30.4 pp, and the true vessel-disjoint protocol—run with ten draws after the temporal numbers were frozen—lands at 0.624 ± 5.4 pp, within 0.6 pp of temporal: two disjoint honest protocols agree with each other and sit 30pp below the leaky one. The reason is structural (Table I): DeepShip's clip-random test clips have an adjacent 5 s window in the training set 89.6% of the time—the two clips are the *same*

TABLE IV

THE PROTOCOL LADDER (MEAN ± DRAW SD; LOG-MEL, CNN_SMALL)

Protocol	OceanShip log-mel	OceanShip log-mel spec	DeepShip log-mel
clip random	$0.367 \pm .004$	$0.365 \pm .004$	$0.922 \pm .006$
recording random	$0.351 \pm .013$	$0.346 \pm .013$	$0.684 \pm .029$
honest (vessel/temporal)	$0.320 \pm .050$	$0.312 \pm .048$	$0.618 \pm .009$
vessel-disjoint (DeepShip, n=10)			$0.624 \pm .054$
clip − honest (paired)	$+5.1 \pm 4.5$ pp (p=.064)	$+5.3 \pm 4.6$ pp (p=.063)	$+30.4 \pm 0.8$ pp (p=.002)

A tag-aware recording-cluster bootstrap (B=2000) over all released draws confirms the ladder and quantifies the allocator effect on OceanShip vessel-disjoint draw SD: 5.0 pp random (n=25), 4.9 count-stratified, 2.6 SGKF, 1.2 clip-mass (n=6)—the same ordering on `cnn_wide` and `logreg`: the allocator a comparison inherits, not the number of draws, decides the observed noise level. DeepShip vessel-disjoint: ten draws over 290 vessel groups (48 AIS-unjoined recordings kept as singletons), every run verified at 0% overlap of recordings and *known* identities (the 48 are guaranteed distinct recordings, not distinct ships—see below); it lands 0.6 pp from temporal ($t95$ $[0.585, 0.663]$)—two disjoint honest protocols, one number—and 30.8 pp below clip random; paired over the three shared seeds the clip−vessel leak is $+28.5 \pm 1.7$ pp ($p = 0.001$). Singleton sensitivity: excluding those 48 recordings (ten draws) shifts the mean to 0.567 (SD 4.4 pp; $t95$ $[0.536, 0.599]$), 5.7 pp below full-corpus (Welch $p = 0.02$)—*consistent with* residual identity leakage, though the exclusion also changes the test composition (29% explained, $p = 0.54$); we report direction and magnitude and disclose the singleton mass rather than claim a causal estimate. Pooled macro-F1: 0.11 ± 1.4 pp (OceanShip vessel), 0.61 ± 5.4 pp (DeepShip vessel). Draw counts differ per column: OceanShip clip/recording 5, vessel 25; DeepShip clip/recording/temporal 3, vessel 10.

audio event—whereas on OceanShip a test clip's nearest training neighbour shares its recording only 5.7% of the time. "We used a random split" is not a comparable specification, and no dataset paper states how its recordings were cut into clips. The DeepShip numbers also calibrate the pipeline against the literature: our honest-protocol accuracy (0.618; pooled macro-F1 0.61 on the harder vessel-disjoint protocol) sits below the best recording-level DeepShip figure we are aware of (0.7226 macro-F1 [16]), with a far smaller model—as expected when moving to a strictly harder protocol; order-of-magnitude agreement, not replication.

B. Decomposing the Draw Variance

Let the draws of a scheme partition vessels into train/validation/test. Each draw d induces a test-set class composition $n_d(c)$ (number of test clips of class c). Holding the model fixed,

$$\mathrm{acc}_d \;\approx\; \frac{1}{N_d} \sum_c n_d(c)\, r(c), \tag{1}$$

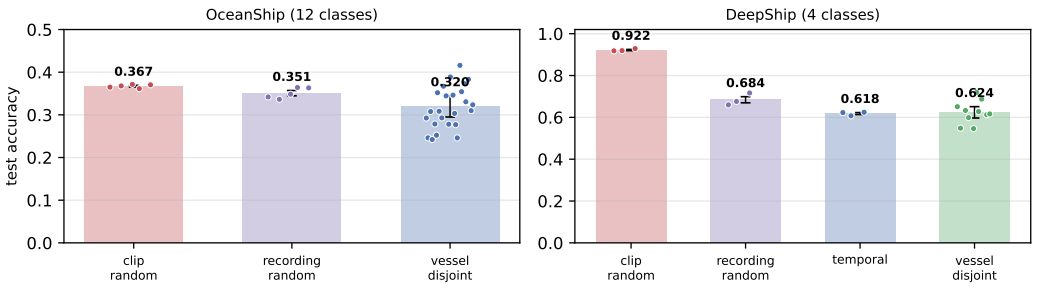


Fig. 3. The protocol ladder on both corpora; each dot is one independent draw, bars mark scheme means (±SD/2). Same code, model, recipe; clip-vs-honest gap: 5.1 pp (paired) on OceanShip, 30.4 / 30.8 pp (temporal / vessel, n=10) on DeepShip; the honest draw spread dwarfs the leaky protocol's.

where $r(c)$ is the per-class recall and $N_d = \sum_c n_d(c)$. We estimate $r(c)$ once, pooled over the draws of the scheme, and predict each draw's accuracy from its composition alone; Eq. (1) would be exact if the per-class recalls were identical across draws; what it ignores is draw-to-draw variation in the recalls themselves, not class interactions. The variance of the predicted accuracies is the *composition component*; the variance of the residuals (observed minus predicted) is the *residual* component—initialization and optimization noise *plus* any remaining draw-level effects (training-set composition, vessel difficulty). It is an empirical explained share, not an identified set of independent variance components, and the estimator differs by row of Table V. For the vessel row (25 draws $\geq$ 12 classes) the composition component is the least-squares projection of the observed accuracies onto the composition shares: residuals are orthogonal to fitted values, so the covariance term vanishes by construction (measured $2\,\mathrm{Cov}(C, E) < 0.01\,\mathrm{pp}^2$) and the components add in quadrature, $4.69^2 + 1.64^2 = 4.97^2$. For the clip and recording rows (5 draws, where OLS is underdetermined) $r(c)$ is the clip-pooled per-class recall measured on the same draws. The clip row is trivially additive ($\mathrm{Var}(C) = 0$); the recording row is not—$2\,\mathrm{Cov}(C, E) = 0.20\,\mathrm{pp}^2$—so its 54% is the variance ratio $\mathrm{Var}(C)/\mathrm{Var}(A)$, not an orthogonal share (cross-terms are footnoted in Table V). At the clip level, where every draw places the same class masses in the test set, this residual is bounded at 0.41 pp, which plausibly reflects initialization and optimization; at the vessel level the residual is larger (1.65 pp) and necessarily contains more than initialization alone.

Table V gives the result on the reference model. The composition component explains **89%**

TABLE V

VARIANCE DECOMPOSITION ON OCEANSHIP (`cnn_small`, LOG-MEL; CLIP/RECORDING 5 DRAWS, VESSEL 25 DRAWS)

Protocol	observed SD (pp)	composition SD (pp)	share (%)	residual SD (pp)
clip random	0.41	0.00	0	0.41
recording random	1.25	0.92	54	0.72
vessel disjoint	**4.97**	**4.69**	**89**	1.65

Composition SD: vessel row = least-squares projection (25 draws, $2\,\mathrm{Cov}(C, E) < 0.01\,\mathrm{pp}^2$); clip/recording rows = fixed pooled per-class recalls (5 draws; recording cross-term $2\,\mathrm{Cov}(C, E) = 0.20\,\mathrm{pp}^2$, so the components are not orthogonal there). The share is $\mathrm{Var}(C)/\mathrm{Var}(A)$ throughout.

of the vessel-disjoint variance over 25 independent draws (composition SD 4.69 of 4.97 pp; permutation $p = 0.0006$), 54% at the recording level, and 0% at the clip level, where the five draws place nearly identical class masses in the test set. Two consequences follow. First, the honest protocol's noise is mostly *metadata*: it can be computed before training and removed by design. Second, Eq. (1) is a falsifiable predictor, which we test next—including in the direction of "nothing to fix" (Sec. VI-D).

C. The Composition Model Predicts Real Experiments

Fig. 5 plots predicted versus observed draw SD for five independent allocation schemes spanning three model families: clip-mass stratified allocation for `cnn_small`/`logreg`/`cnn_wide`, count-stratified allocation, and `StratifiedGroupKFold` [8]. All five lie on or near the identity line ($1.12 \to 1.24$, $0.70 \to 0.94$, $0.92 \to 1.36$, $4.85 \to 4.85$, $2.28 \to 2.60$ pp). The predictor consumes the splits' class composition and one pooled per-class recall table; it never sees the training outcomes of the schemes it predicts. Calibration and prediction are disjoint by construction: the recall table is estimated on the reference scheme's own random draws and frozen, and the schemes it scores were drawn *after* that table was frozen (Sec. IX-C). The residual gap (observed slightly above predicted) is consistent with the residual draw-level variability quantified in Table V. In Sec. VI-D the same model makes a *null* prediction on a second corpus—that there is nothing to fix—which is also confirmed. Out-of-sample honesty, audited by leave-one-draw-out prediction on the 25-draw vessel pool: fitting on 24 draws and predicting the held-out draw from its composition alone gives $r = 0.80$ (permutation $p = 0.0002$,

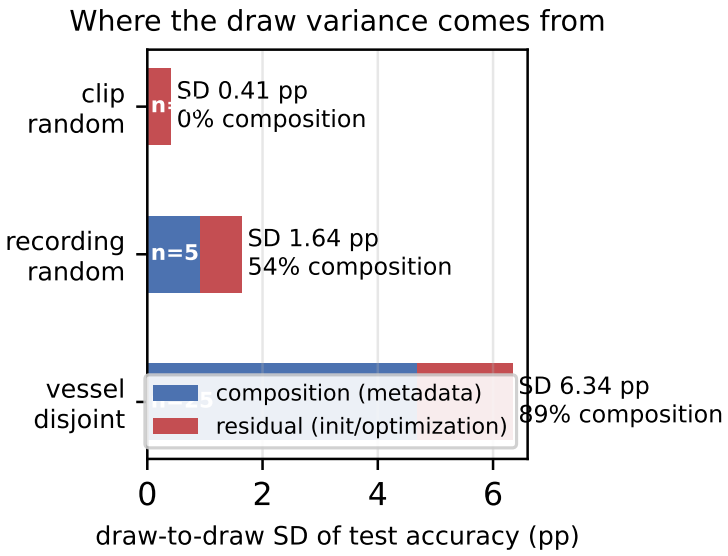


Fig. 4. Out-of-sample anatomy of the decomposition of Table V (25 vessel-disjoint draws). (a) Per-draw accuracy versus the draw's Towing clip-mass share (Pearson $r = +0.71$) with the fitted relation: composition is the mechanism, visible draw by draw. (b) Leave-one-draw-out predicted versus observed accuracy ($r = 0.80$, RMSE 3.1 pp): the composition model predicts accuracies of draws it has never seen.

RMSE 3.1 pp) and LODO $R^2 = 0.59$ against the in-sample 0.89: per-draw accuracy is predictable for unseen draws at ± 3 pp, while the 89% share is an in-sample empirical variance share of the current 25 draws; we report both to keep the in-sample optimism visible.

VI. A Metadata-Only Remedy: Class-Wise Clip-Mass Allocation

A. The Allocator

Given the decomposition, the fix is to allocate whole vessels so that each class contributes its quota of *clips* to test and validation. Per class c with per-vessel clip counts m_v: order vessels by a hash of (draw seed, class, vessel); accumulate vessels into the test set until $\sum_v m_v \geq q_c$ with $q_c = 0.15 \sum_v m_v$; repeat into validation; the remainder is training. No clip crosses a vessel boundary, and because the last vessel admitted may overshoot the quota, each class's test clip count is constrained to within one vessel of the target rather than exactly equal to it—and thus dampens the random deviations of test-set class proportions from the corpus distribution.

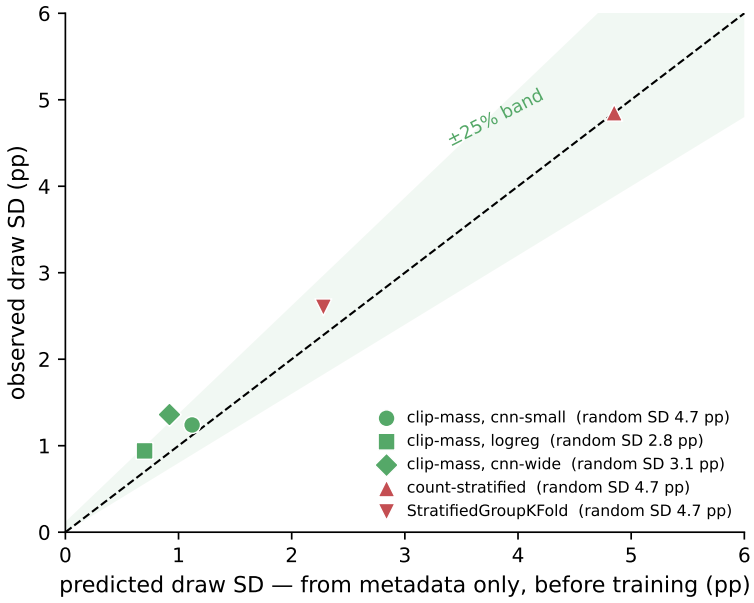


Fig. 5. Predicted vs. observed draw-to-draw SD for five allocation schemes (three model families); shaded band $\pm 25\%$. The calibration-based model tracks the identity line without seeing any training outcome of the schemes it predicts; the legend gives each scheme's random-draw SD.

B. Results: The Variance Collapses to the Observed Floor

Table VI gives the pre-specified comparison. On the reference model, random draws give 0.3157 ± 4.69 pp (5 draws; the released 25-draw pool, Sec. V, agrees at 0.3198 ± 4.97 pp) and clip-mass draws 0.3215 ± 1.24 pp: variance ratio 14.4 (F-test $p = 0.012$; exact permutation $p = 0.004$; 16.1 against the 25-draw random pool)—down to the level of the residual draw-to-draw variability quantified in Table V (1.65 pp; not pure initialization noise, which the clip-level pipeline bounds at 0.41 pp, but the residual band the allocator cannot remove). Because the test sets are now composition-matched, the random, clip-mass, and SGKF schemes have comparable means (within 1.3 pp, Welch $p \geq 0.59$): the allocator removes noise, not difficulty; count-stratified shows a larger mean shift and is discussed separately. Translating SD into experiment design: the draws per system needed to resolve an improvement δ at $\alpha = 0.05$ and power 0.8, computed with the non-central t distribution, are plotted in Fig. 6 (right). At $\delta = 2$ pp: **98** draws for the random scheme, 28 for SGKF, **8** for clip-mass (the normal approximation gives 97 / 26 / 6—it overstates power at small n). A week of GPU budget buys a defensible claim under the remedy and nothing under the convention. These are *unpaired* budgets (one model, draws as units). When

TABLE VI

REMEDY VS. ALTERNATIVES, OCEANSHIP VESSEL-DISJOINT, REC2 RECIPE

Backbone	Scheme	n	mean	SD (pp)	pred. SD	ratio
logreg	random draws	5	0.280	2.76	—	—
	clip-mass	6	0.286	**0.94**	0.70	0.34
cnn_small	random draws	5	0.316	4.69	—	—
	clip-mass	6	0.322	**1.24**	1.12	0.26
cnn_wide	random draws	5	0.304	3.07	—	—
	clip-mass	6	0.312	**1.36**	0.92	0.44
cnn_small	count-strat.	6	0.370	4.85	4.85	0.93
	Strat. GroupKF	6	0.309	2.60	2.28	0.36[†]

[†]variance ratio 3.25; F-test $p = 0.23$; permutation $p = 0.084$—clip-mass shows *observed* lower variability; superiority over SGKF is not established at conventional significance at these sample sizes. All schemes share the same 70/15/15 clip budget and vessel exclusivity; no vessel carries conflicting labels (5 of 892 have minority-class clips, 1.1% of clips; allocation uses the vessel's majority class). Clip-mass variance ratios 14.4 / 8.7 / 5.1; permutation $p \leq 0.013$ (exact enumeration of all $\binom{11}{5} = 462$ scheme-label assignments, statistic = log variance ratio, two-sided; script `w4h_exact_perm.py` in the release).

two systems are evaluated on the *same* draws, the composition term largely cancels in the paired difference: on the same five random draws the `cnn_small−cnn_wide` difference has SD 1.9 pp (against 5.6 pp if the two models responded to composition independently), and under clip-mass allocation the paired-difference SD falls to 0.6–1.4 pp. At small n the normal approximation overstates power, so all required-N figures in this paper—independent and paired—are non-central-t planning estimates: SD 1.9 pp requires $n = 10$ paired draws (2.3 pp: 13), clip-mass 0.6–1.4 pp requires $n = 3$–7, and on the independent two-sample design of Fig. 6 (right) the schemes need 98 / 28 / 8 draws per system. The pattern replicates across three model pairs (the `cnn_wide−logreg` paired SD is 0.9 pp against 4.1 pp independent) and both allocation schemes. These SDs rest on 5–6 draws each (the single-model clip-mass SD's 95% interval spans 0.8–3.0 pp), so the budgets are planning estimates, not measured costs. Pairing and the allocator are complementary, not alternatives: pairing removes the shared composition term, the allocator removes what remains. Across the twelve hypothesis tests reported in Tables IV–VI, a Benjamini–Hochberg correction at FDR 0.05 changes no conclusion; the test family and raw p-values are recoverable from the released per-draw files via the aggregation scripts.

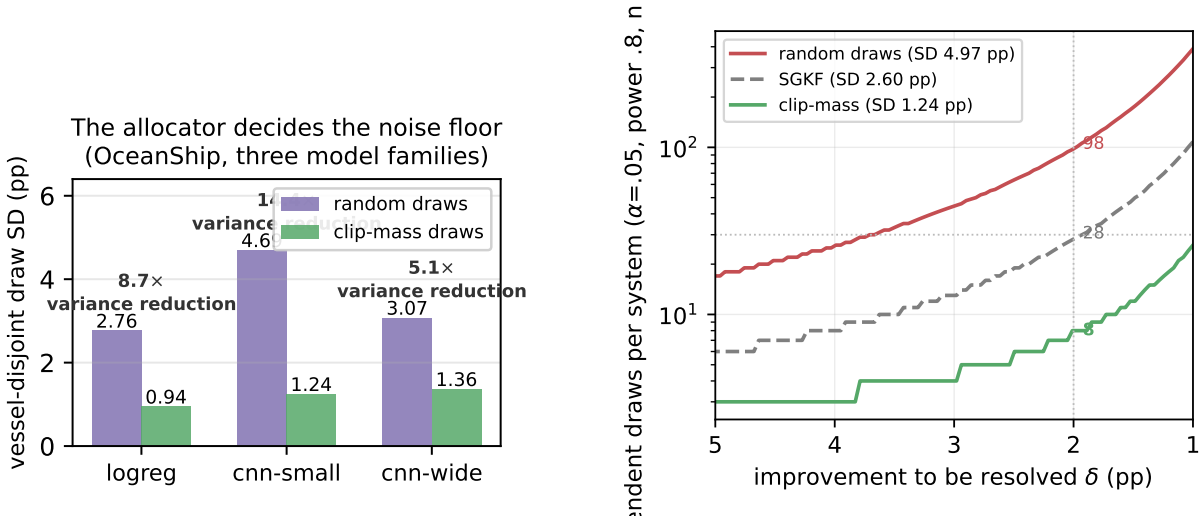


Fig. 6. Left: Table VI at a glance—the vessel-disjoint draw SD under random versus clip-mass allocation, per model family (ratios annotated); the allocator, not the backbone, decides the observed noise level. Right: independent draws per system required to resolve an improvement of δ pp ($\alpha = 0.05$, power 0.8, two-sample non-central t), for the random convention, `StratifiedGroupKFold`, and clip-mass allocation; dashed line: a realistic weekly GPU budget. Planning estimates under SDs measured on 5–25 draws.

C. Cross-Model Generalization and Two Instructive Failures

The collapse holds on every model family tested (Table VI, Fig. 6): a linear probe (variance ratio 8.7, exact permutation $p = 0.013$), the reference CNN (14.4, $p = 0.004$), and the wider CNN (5.1, $p = 0.009$); per-model SD ratios are 0.34, 0.26, 0.44, all under the pre-specified 0.6 bar. Two failures sharpen the mechanism. *Count-stratified* allocation, balancing the number of *vessels* per class, leaves the variance unchanged (ratio 0.93) with a $+5.4$ pp mean shift (Welch $p = 0.096$, not significant—we do not read its direction): vessels carry 3–250 clips each (an $80\times$ range), so count balance does not balance information mass. *StratifiedGroupKFold* [8], the standard tool, halves the variance (ratio 3.25) but remains $2.4\times$ worse than clip-mass. In the experiments reported here, stratification had to act on identity-*mass*, not identity counts—and the composition model predicted both outcomes in advance.

D. A Low-Composition-Variance Control on DeepShip

The composition model makes a falsifiable cross-corpus prediction. DeepShip's four classes are balanced and its 609 recordings are numerous relative to the class count, so a random recording-level draw barely moves the test composition: predicted composition SD 0.30 pp—an order of

TABLE VII

LOW-COMPOSITION-VARIANCE CONTROL: DEEPSHIP REC_RANDOM (CNN_SMALL, LOG-MEL)

Scheme	n	mean	SD (pp)	pred. SD (pp)
random draws (frozen)	5	0.699	2.95	0.30
clip-mass draws	6	0.654	3.92	

magnitude below OceanShip's 4.3 pp. Stratification should therefore buy nothing. Consistently, no variance-reduction benefit is detected under the tested settings (Table VII): random draws 0.699 ± 2.95 pp, clip-mass draws 0.654 ± 3.92 pp, ratio 1.33 (inside the pre-specified $[0.6, 1.67]$ band; the -4.5 pp mean shift is not significant, Welch $p = 0.06$—and with 5–6 draws per arm this design cannot establish equivalence, only fail to detect a benefit). Two further observations strengthen the attribution. First, the frozen splits reproduce month-earlier values on the same protocol to all reported digits. Second, most of DeepShip's residual draw variance ($\sim$3 pp) is *not* composition (the model explains only $\sim$10% of it)—it reflects which individual ships are held out. The predictor is thus discriminative in both directions: it identifies where the remedy applies (OceanShip, every model family) and where it does not (DeepShip), before any training run. The ten-draw vessel-disjoint protocol (Table IV) completes the picture on both sides: DeepShip's identity leak is real ($0.922 \rightarrow 0.624$) even though its composition remedy shows no benefit, and the stricter protocol's draw variance, now measurable with ten draws, is likewise not composition-shaped (21.9% explained, permutation $p = 0.66$)—the null control holds at the vessel level too. Removing the leak and stabilizing the remaining variance are two different interventions, and only the second was ever claimed to be unnecessary on DeepShip. The same ten draws give DeepShip's four-class structure its honest denominator. Pooled per-class recall under vessel isolation: Tanker 0.71 ± 8 pp, Passenger 0.66 ± 8 pp, Tug 0.64 ± 10 pp, Cargo 0.52 ± 10 pp—a 19 pp spread in which every class stays far above the 0.25 chance rate (pooled macro-F1 0.61 ± 5.4 pp), against OceanShip's three-tier collapse under the same protocol (best class 0.75; nine classes ≤ 0.07; four or five never predicted). Class *granularity*, not corpus identity, decides whether honest evaluation strands classes: four balanced classes over 609 recordings leave no tail; twelve AIS-administrative classes over 892 vessels leave nine.

TABLE VIII

DIAGNOSTIC ANALYSES AND TESTED INTERVENTIONS FOR TAIL-CLASS COLLAPSE

Intervention	Pre-registered test and outcome
Capacity (ResNet-18, 11.2M)	memorizes the training set (0.9999), tests at 0.378; $+1.0$ pp over `cnn_small` [16].
Post-hoc prior adjustment	offline logit adjustment on saved probabilities upper-bounds decision-rule fixes that leave training untouched (it cannot alter learned representations, as in-training loss re-weighting could): best $+2.0$ pp macro-F1 (accuracy -2.3 pp).
Selective prediction	accuracy–coverage curves are flat for three confidence scores (plots in the release): acc@50% $= 0.367$ ($+4.9$ pp); a head-only system still answers 97.3% of tail clips.
Training-mass analysis	diagnostic dose–response: recall tracks training clip count ($\rho = +0.83$) better than vessel count ($\rho = +0.68$); 42-vessel Passenger at 0.000 recall. Identity floor confirmed only for the 2 classes with < 5 training vessels (pre-specified, hit).
Frozen pretraining	AudioSet AST [17] + linear probe on all 53,770 clips: tail top-2 0.116 vs. 0.114 for the small CNN.
Full fine-tuning	end-to-end AST (2 seeds $\times$ 8 epochs): the only tested remedy that moves macro-F1 ($0.108 \rightarrow 0.142$), but tail top-2 reaches 0.178 (GO bar 0.30); val peaks at epoch 2 then degrades to train-acc 0.81.

VII. LIMITS OF TESTED INTERVENTIONS: THE EFFECTIVE CLASS STRUCTURE

A. Tested Interventions and Diagnostics

Under the honest protocol the reference model collapses to three classes (Towing 0.72, Cargo 0.58, Tug 0.37 recall; nine classes ≤ 0.07, four to five never predicted; macro-F1 0.11 against a 0.083 uniform baseline). For every obvious explanation we pre-specified a discriminating experiment and ran it (Table VIII).

B. A Three-Tier Tail, Predictable from Metadata

The fine-tuning gradient is not uniform (Fig. 7): Fishing (top-2 $0.296 \rightarrow 0.442$) and Pleasure Craft ($0.208 \rightarrow 0.339$) are partially rescued; Passenger does not move; Search-and-Rescue and Tanker stay at zero; head classes are squeezed (Cargo recall $0.58 \rightarrow 0.32$). Combined with the

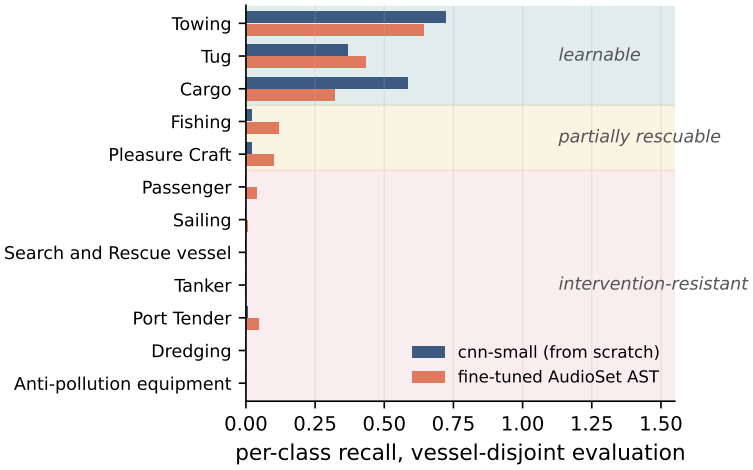


Fig. 7. Per-class recall under vessel-disjoint evaluation: from-scratch CNN vs. fully fine-tuned AudioSet AST. Shaded tiers: learnable / partially rescuable / intervention-resistant. Fine-tuning partially rescues the high-mass small-boat classes, leaves Passenger unchanged, cannot revive SAR and Tanker, and trades away head-class recall.

identity floor (Fig. 8, Table IX) and the confusion structure (Fig. 9)—tail classes project 70–80% of their mass onto Towing, Tanker projects 60% onto Tug, Tug itself 37.8% onto Cargo, and the class means in feature space score *below* chance—the tail resolves into three tiers: *learnable* (Towing, Cargo, Tug; 3 classes, 61.5% of clips), *partially rescuable* (Fishing, Pleasure Craft), and *intervention-resistant* (Passenger, Sailing, SAR, Tanker, and the identity-floor classes Port Tender, Dredging, and Anti-pollution equipment—a collapse we predicted before running and confirmed). "Resistant" means not recovered by any of the tested interventions; it is not a claim that no intervention could ever work. The cross-corpus contrast confirms the attribution: the same pipeline learns all four DeepShip classes under the honest protocol (recalls 0.55–0.75), so the observed contrast is consistent with a substantial role of the benchmark's label granularity, though it does not isolate granularity as the sole causal factor; a within-corpus class-merge experiment is the clean test and remains future work.

VIII. A Reporting Recipe

The audit reduces to five reportable numbers, all computable without new modelling: (1) the allocation unit, and how the corpus slices recordings into clips (tiling ratio, clips per recording, stream-position availability); (2) the number of independent draws and the draw-to-draw SD per scheme; (3) the composition share of that SD via Eq. (1); (4) the required-N table for the

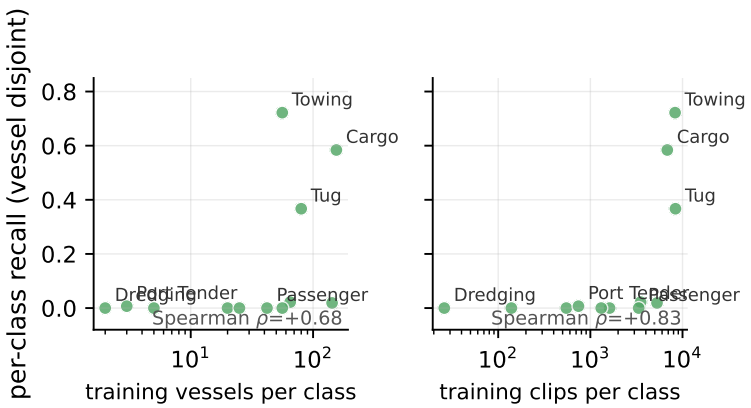


Fig. 8. Dose–response of per-class recall against training-set identity diversity (log scale). Recall tracks training *clips* ($\rho = +0.83$) better than training *vessels* ($\rho = +0.68$): identity scarcity floors the two < 5-vessel classes but is not the main bottleneck.

TABLE IX

PER-CLASS METADATA AND OUTCOME (VESSEL-DISJOINT, POOLED 5 DRAWS)

Class	recall	train vessels	train clips
Towing	0.722	56	8,306
Cargo	0.584	155	6,837
Tug	0.367	80	8,382
Fishing	0.022	65	3,510
Pleasure Craft	0.019	143	5,287
Port Tender	0.007	3	743
Passenger	0.000	42	3,346
Sailing	0.000	46	1,099
Search and Rescue	0.000	15	318
Tanker	0.000	13	112
Anti-pollution	0.000	5	18
Dredging	0.000	2	26

Passenger (42 vessels, 3,346 clips) refutes vessel count as the bottleneck; Port Tender and Dredging confirm the pre-specified identity floor.

claimed effect size (Fig. 6 (right)); (5) the effective class structure under the honest protocol (per-class recall and top-2, never-predicted classes, < 5-identity floors). We release the frozen splits, the allocator, and the required-N calculator so that this is a five-minute post-processing step rather than a re-review.

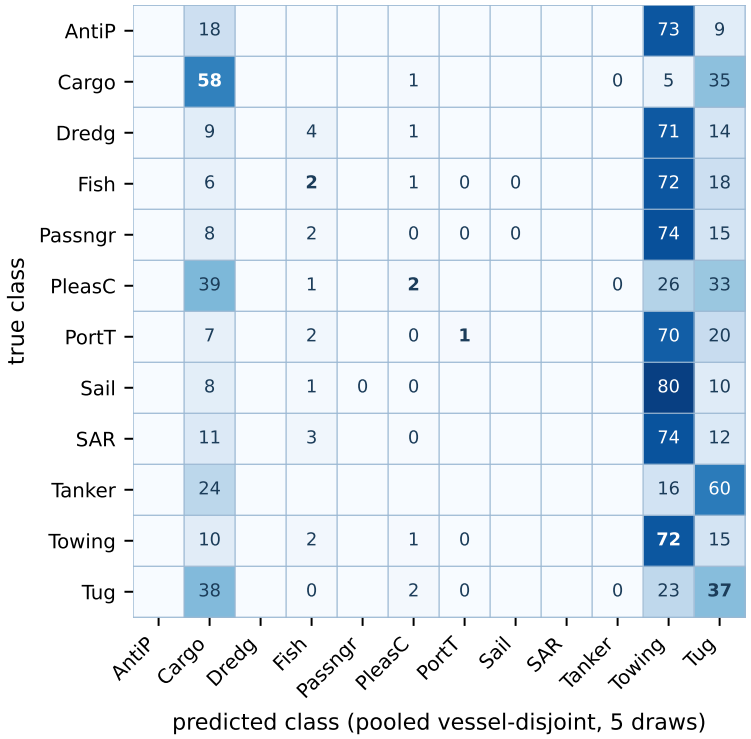


Fig. 9. Row-normalized confusion matrix (pooled over five vessel-disjoint draws). Tail-class mass flows almost exclusively into the three head columns; within the tail, confusion is diffuse. Under the honest protocol the 12-class benchmark behaves as a 3-class problem.

IX. Discussion and Limitations

A. Relation to Existing Guidance

Published guidance says *what* to do (use recording- or identity-level splits) and *why* (adjacent clips share noise and propagation conditions [1]); it does not say what the change does to the statistics of an evaluation. Our numbers supply the missing half: the honest protocol multiplies draw noise by $12\times$, that noise is 89% composition (59% predictable out-of-sample) and therefore largely foreseeable before training, and the remedy makes rigorous comparison affordable ($\sim$100$\rightarrow$8 unpaired draws per system; measured on real model pairs, pairing alone cuts the requirement to 10–13 under the non-central-t power calculation—the paired-difference SD is 1.9–2.3 pp, not the 5.6 pp independence would predict—and adding the allocator cuts it again to 3–7). We also learned the aggregation lesson the hard way: an unfiltered pool of 25 vessel-disjoint runs of mixed provenance—six allocation schemes and two probes, a superset of and distinct from the 25 random draws of Table IV—has a draw-to-draw SD of 8.2 pp, which pools

between-scheme variance with draw variance. Re-aggregated per scheme (released scripts), the same runs reproduce 5.0 pp under the random allocator and 1.2 pp under clip-mass: tag-aware aggregation is not optional bookkeeping but the difference between a statistic and a mixture. This reframes identity-disjoint evaluation from "a stricter convention" into "a protocol with known, managed statistics"—the same transition clinical research underwent when multi-centre variance moved from anecdote to design parameter.

B. What the Remedy Does and Does Not Claim

Clip-mass allocation does not make models better; it makes comparisons cheaper and honest numbers stable. The $4.69 \rightarrow 1.24$ pp collapse makes a two-draw comparison feasible where it would previously have been dominated by composition noise—though two observations still support only weak inference—and paired designs (same draws for all systems) compose with it for a further measured 3–$4\times$ reduction in the number of draws (Sec. VI). The remedy also dampens the draw-to-draw random deviations of test-set class proportions from the corpus distribution; a random vessel draw can make the test set systematically easier or harder than the corpus, and averaging over few draws does not fix a biased draw.

C. Threats to Validity

Internal. All comparisons share one frozen split family and one recipe; run identifiers encode recipe and protocol, and every run's final learning rate is logged and checked. The recall table feeding the predictor was frozen before the scored schemes' draws existed (Sec. V-C). *External.* The remedy is validated on one corpus's honest protocol across three model families, with the null control on a second; a third corpus with intermediate composition variance would complete the dose–response. Both benchmarks also ultimately draw on recordings from the same regional collection network (Ocean Networks Canada, Strait of Georgia)—on disjoint vessels, recording campaigns, and periods—so the null control demonstrates that the composition *mechanism* generalizes across corpora, not that the recording conditions are independent; a corpus collected outside this network remains the missing external point. Absolute accuracies are low (0.24–0.37); our claims concern *variance and structure*, which we argue are protocol properties, and the capacity/pretraining interventions bound the role of model choice. A SOTA-scale system would shift the per-class recall table feeding the predictor—but the predictor is re-estimated per system from its own random draws, so the framework, not the constants, is

the contribution. *Construct.* Equation (1) holds the per-class recalls fixed: it cannot capture draw-specific recall variation arising from training-ship identity, test-ship difficulty, or acoustic conditions; class interactions, if present, are one possible contributor among these. The DeepShip residual (composition explains only $\sim$10% of a still-substantial 3 pp spread) makes this visible. The 85% share of the pre-specified five-draw analysis was re-estimated independently on the released 25-draw pool (89.0%, permutation $p = 0.0006$; Table V); the remedy pools (n=6 per scheme) and the DeepShip vessel pool (n=10: 21.9%, $p = 0.66$, consistent with the null control) remain too small for their own decomposition—the required-N analysis of Sec. V applies to our own confirmatory pipeline as much as to anyone's. Fine-tuned AST ran with two seeds at a fixed budget; the tail conclusion rests on the pre-specified bar rather than an exhaustive sweep. *Labels.* OceanShip's AIS categories are administrative fishing-vessel, port-tug, rescue-boat classes, not acoustic families. Our results quantify that mismatch—the effective class structure has three tiers, and the metadata predicts which classes fall in which—but do not claim it is universal. Benchmarks built from operational metadata should expect exactly this analysis.

X. Conclusion

Honest evaluation in UATR is currently a trade with unquantified terms: identity-disjoint splits remove a real leak and, on OceanShip, replace 0.4 pp of clip-level noise with 5.0 pp of metadata-driven noise that single-draw papers silently absorb. We showed this noise is 89% composition (59% predictable out-of-sample), removable by a class-mass allocation to the lowest variability observed across the tested allocations ($14.4\times$ variance reduction, exact permutation $p = 0.004$), general across model families, and correctly predicted to be unnecessary on DeepShip. We also showed what the resulting honest numbers mean: a three-tier effective class structure that the tested interventions—including full fine-tuning of an AudioSet-pretrained transformer—cannot open beyond its predicted limits under the tested settings. Benchmarks should report the allocation scheme, the draw variance, and the effective class structure; without them, "accuracy on OceanShip" is not one number but a distribution.

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REPRODUCIBILITY

Frozen splits (SHA-256-seeded, content-verified at load), all training and evaluation code, the allocation scripts, per-clip predictions for every run, the required-N calculator, a run manifest (`run_manifest.csv`: one row per released draw with corpus, model, protocol, allocation scheme, split seed, and metrics), and the machine-readable aggregation outputs (`w3e_bootstrap.json`, `w3e2_variance.json`, `w3e3_lodo.json`) are released at https://github.com/gzhgzy1234567890-lang/uatr-honest-evaluation (a DOI-minted archive follows at acceptance). Every table regenerates from the released per-draw files. On DeepShip, the 48 recordings whose AIS join failed enter the vessel-disjoint protocol as singleton vessels (group key `singleton::<rec_key>`); the runner's leak check applies the same rule, so what the split promises and what it verifies are identical.

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